CSI3005 Case Study Presentation Master Script

Student: **Monasree S (23MIC0084)** | Slot: E2 | Evaluation Target: **20 / 20 Marks**

Cheat-Sheet: 10-Min Viva Defense

⏱️ Step 0: The 30-Second Opening Hook (Say This First!)

OPENING HOOK

> 🗣️ **EXACT SCRIPT TO SAY:**
> "Good morning Professor. My project is **'Beyond Crash Counts: Spatiotemporal Visual Analytics for Urban Road Safety'**. In NYC, citywide collision counts have dropped by 56%, which municipal leaders celebrated. But my visual analysis discovers a critical hidden danger: **the probability of a crash causing injury or death actually DOUBLED from 19.3% to 40.0%**! Today I will demonstrate how spatiotemporal visualization in R decodes where and why this severity inversion is happening."

📁 Component 1: Dataset Selection & Data Preparation (5 Marks)

5 / 5 MARKS

2.27 Million	14 Years	488,444	13 Core
RAW COLLISIONS	JULY 2012 – JUNE 2026	BOROUGHS RECOVERED	STANDARDIZED FACTORS

> 🗣️ **WHAT TO SAY TO THE FACULTY:**
> "I used the official **NYPD Motor Vehicle Collisions Public Registry** from NYC Open Data (Dataset ID: `h9gi-nx95`). It has 2,269,187 real rows and 29 columns across 14 years. I implemented an 8-step data preparation pipeline in R:"

- **Step 1: Schema Normalization:** Cleaned all 29 column names to `snake_case` using `janitor`.
- **Step 2: Temporal Decomposition:** Extracted crash year, month, hour (0–23), and weekday (Mon–Sun).
- **Step 3: Bounding-Box Filtering:** Filtered to verified NYC coordinates `[40.48–40.93 N, -74.28 to -73.68 W]`, discarding 7,591 Null Island (0,0) noise points.
- **Step 4: Spatial Borough Recovery (The Highlight!):** 691,375 records lacked borough names. Using point-in-polygon spatial imputation with NYC shapefiles in package `sf`, I **recovered 488,444 valid geocoded crashes**!
- **Step 5: Casualty Imputation:** Replaced missing injuries/deaths with 0 per NYPD reporting rules.
- **Step 6: Severity Tier Factor:** Created a mutually exclusive 3-level factor: `Fatal Crash`, `Injury Only`, and `Property Damage`.
- **Step 7: VRU Composite Metric:** Combined pedestrian and cyclist trauma into Vulnerable Road Users (VRU).
- **Step 8: Behavioral Mapping:** Vectorized 67 messy police strings into 13 standardized driver factors.

> 🎓 **Killer Academic Defense Point (Methodological Safeguard):**
> "Professor, I explicitly enforce an **Exposure Denominator Safeguard**: Raw crash counts indicate volume exposure (traffic density), not necessarily bad road geometry, because we lack vehicle miles traveled (VMT). Therefore, I analyze *conditional severity rates (% injury/fatal)*, never raw volume alone!"

Visual Selection, Design Critique & Streaming Math

Components 2 & 3: Theory Defense, Tufte Lie Factor, and Patent Architecture

Components 2 & 3 • 10 Marks

📊 Component 2: Visualization Selection & Cognitive Defense (5 Marks)

5 / 5 MARKS

> 🗣️ **WHAT TO SAY ABOUT VISUAL TECHNIQUES:**
> "I defended all visual choices using **Cleveland & McGill's graphical perception hierarchy**, which proves humans perceive position along common scales and length with the highest cognitive accuracy, while area and color hue have higher error rates."

- **Figure 1: 2D Hexagonal Spatial Binning (`stat_bin_hex` + `viridis magma`):**
  Why not scatter dots? 2 million scatter points cause severe overplotting—the entire city becomes an undifferentiated black blob. Hexagons tessellate space uniformly without directional distortion, encoding collision density into continuous logarithmic color luminance. Reveals acute hotspots along **Midtown Manhattan (FDR Drive), Queens Blvd, and Cross Bronx Expressway**.
- **Figure 2: Dual-Panel Longitudinal Multi-Series Time Series (`geom_line` + LOESS):**
  Why dual-panel? Exploys position along an aligned common scale. Top panel tracks volume (0–22,000); bottom panel tracks % injured (10%–50%). Decoupling volume from severity clearly exposes the structural break during COVID-19.
- **Core R Packages:** `ggplot2` , `sf` , `hexbin` , `viridis` , `data.table` , `plotly` , `shiny` , `bs4Dash` .

🎨 Component 3: Design Critique, Tufte Lie Factor & Patent Diagram (5 Marks)

5 / 5 MARKS

> 🗣️ **WHAT TO SAY ABOUT TASK 3 (CRITIQUE & REDESIGN):**
> "In Task 3, I critiqued an intentionally deceptive bar chart used in public-sector reporting and applied Edward Tufte's mathematical Lie Factor formula to correct it:"

- **Flaw 1: Truncated Y-Axis (Lie Factor = 4.20x):** The chart cut the vertical axis at 3,100 collisions instead of zero. Because bar charts encode value by bar length, this exaggerated a routine 21.0% seasonal change into an apparent 88.1% visual collapse! $$ ext{Lie Factor} = rac{ ext{Size of effect in graphic}}{ ext{Size of effect in data}} = rac{88.1\%}{21.0\%} = 4.20$$ A Lie Factor of 4.2x severely deceives civic leaders into declaring a false public-safety catastrophe!
- **Flaw 2: Red-Green Emotional Bias:** Alternating red/green induced fear and violated WCAG 2.1 AAA accessibility for colorblind individuals (8% of males).
- **The Redesign (Figure 4):** Restored a zero baseline, applied corporate slate blue (`#2B6CB0`), direct value labels, and maximized Tufte's data-ink ratio.

> 🗣️ **WHAT TO SAY ABOUT FIGURE 5 (PATENT-GRADE ARCHITECTURE):**
> "To satisfy Task 4's changed requirement for high-frequency streaming data, I designed a **patent-grade 10-module closed-loop telemetry architecture**. It features HTTP/2 CAD ingestion (120 events/sec), Kafka borough-partitioned ring buffers, Flink 15-minute tumbling and 24-hour sliding windows, Redis sub-5ms serving, and an automated Z-score anomaly trigger: $$Z_t = rac{ ext{Rate}_{ ext{inj}}(t) - \mu_{24 ext{h}}}{\sigma_{24 ext{h}}} \ge 3.00 \quad (p < 0.0013)$$ with a Poisson rate safeguard ($N \ge 15$) to prevent false alarms during sparse hours."

Key Findings, RStudio Code & Shiny Live Demo

Component 4 & Live Demonstration Walkthrough: The Final 20/20 Grade

Component 4 & Live Demo

💡 Component 4: 4 Major Findings & 4 Policy Recommendations (5 Marks)

5 / 5 MARKS

+20.8%	4.83x	131.0	117.9s
POST-COVID INJURY SURGE	VRU LETHALITY DISPARITY	SPEED DEATHS / 10K CRASHES	FULL R PIPELINE TIME

- **Finding 1: Post-COVID Severity Inversion:** Lockdown crashed monthly volume by -77.9% (from 18.6k to 4.1k/mo). But empty roads led to speeding, causing injury probability to double from **19.3% to 40.0%**!
- **Finding 2: Vulnerable Road User (VRU) Disparity:** Motorists suffer 2.71 deaths per 1,000 injuries, while pedestrians suffer **13.10 deaths per 1,000 injuries (4.83x deadlier!)**, accounting for half of all deaths (1,805).
- **Finding 3: Speed Kills:** Unsafe speeding causes 131.04 deaths per 10k crashes (**12.7x deadlier** than inattention).
- **Finding 4: Spatial Clustering:** Collisions concentrate on multi-lane arterials (Queens Blvd, Atlantic Ave).
- **4 Strategic Policies:** (1) 24/7 automated radar cameras; (2) Intersection daylighting (20-ft setbacks); (3) Concrete bike lane barriers; (4) Automated spatial reverse-geocoding for open-data integrity.

💻 Live RStudio & Shiny Demo: Exact Cues to Show the Faculty

LIVE DEMO GUIDE

🖥️ STEP 1: RUN THE PIPELINE IN RSTUDIO CONSOLE

Type: `source("run_pipeline.R")`

Say to faculty: "Professor, notice the console: using `data.table`, it ingests all 2.27 million rows in just **5.45 seconds**. It executes all 7 pipeline steps, evaluates **8 automated unit test suites with 42 passing assertions**, and completes the entire end-to-end run in **117.9 seconds**."

🚀 STEP 2: LAUNCH THE SHINY DASHBOARD

Type: `shiny::runApp("app")` → It opens on `http://127.0.0.1:6176`.

Point your mouse and explain each plot:

- **1. Top KPI Ribbon:** 2,269,187 Total Crashes | 756k Injured | 3,617 Killed | 24.62% Injury Rate | 58.0% VRU Share.
- **2. Top-Left Plot (Monthly Volume):** "Point at March 2020: clearly see the sharp COVID drop and post-lockdown plateau."
- **3. Top-Right Plot (7x24 Diurnal Heatmap):** "Hover over Friday 16:00–17:00 (peak 26k volume); contrast with Sunday 3 AM (low volume, peak fatal crash rate)."
- **4. Bottom-Left Plot (Borough Trends):** "Shows annual crash trajectories across all 5 boroughs from 2013 to 2025."
- **5. Bottom-Right Plot (Factor Dumbbells):** "Compares volume vs fatality rate: Driver Inattention is high volume but low lethality; Unsafe Speeding is extreme lethality (131 deaths/10k)."
- **6. Interactivity Demo:** "Slide the year slider to '2020–2024' or select 'Brooklyn' from the dropdown: all plots and KPIs instantly re-render in sub-second time via reactive expressions!"

⚡ Expected Faculty Questions & 1-Sentence Winning Answers:

Q1: Why did injury rates double after COVID?

Answer: "Lower traffic congestion allowed higher vehicle operating speeds, increasing kinetic energy on impact."

Q2: Why use hexagonal bins instead of a heat map or scatter plot?

Answer: "Scatter plots suffer from severe overplotting with 2M points; hexagons avoid directional bias and tessellate uniformly with minimal perimeter-to-area distortion."